

Course Project: Big Data Concepts and Implementations

Big Data Processing and Visualization of NYC Taxi Trends

Submitted By:

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Introduction

With cities growing and data being generated faster than ever, it's clear that big data plays a major role in how we understand urban life. Transportation, in particular, produces a ton of useful information, such as when and where people travel, how they pay, and how long they stay on the road. If we can make sense of it, we can learn a lot about how a city moves. But working with real-world data is not always smooth. NYC taxi data, while rich in detail, is also messy and unstructured at times. There are trips with zero distances, missing tip values, odd timestamps, and even strange fare combinations. Cleaning and preparing this data is a critical first step before any kind of analysis can happen. It's a good reminder that in the world of big data, half the work is often just wrangling the data into a usable form.

In this project, I explored New York City's Yellow Taxi trip data for the entire year of 2019. January was specifically used for running the PySpark job on Dataproc, as the full dataset exceeded the memory limitations of our n4-standard-2 cluster configuration. The goal was to spot useful patterns in how people travel: when rides peak, which zones are busiest, and how tip amounts relate to trip distance. I used tools like BigQuery for fast querying, Dataproc with PySpark for distributed processing, and Looker Studio to create clean dashboards. Altogether, this became a full pipeline that starts with messy records and ends with something useful and visual.

Background

I chose to work with this dataset because transportation is one of those areas where data speaks volumes about how a city functions. Things like where people travel, when they move, how long the trips last, and what kind of payment is used these all give clues about how busy certain areas are, when demand peaks, and how the city flows throughout the day.

New York City felt like the best place to analyze this. It's not just any city, it's one of the most active and diverse urban hubs in the world. With millions of taxi trips recorded in just one year, the data is huge, and that gave me the perfect chance to work with tools built for big data. That made this dataset a really solid choice for practicing real-world data engineering and analysis.

Methodology

Accessing and Exploring the Dataset in BigQuery

The first step in this project was connecting to the NYC Yellow Taxi trip dataset, which is publicly hosted on Google BigQuery. This dataset contains over 100 million trip records from 2019, including details like pickup and dropoff times, fare amounts, tip amounts, distance traveled, and location IDs.

One reason this dataset was a great fit is its size. The full 2019 data is roughly 100 GB—too large to store or process efficiently on a local machine. Since it was already hosted in BigQuery, I could start querying it right away without worrying about downloading or managing raw files. That allowed me to jump straight into exploration and analysis.

I began by writing basic SQL queries in BigQuery to understand the dataset structure and look for common data issues. Some records had zero distance or fare amounts, while others were missing timestamps or had inconsistencies. These checks helped prepare the data for filtering and cleaning in the next step.

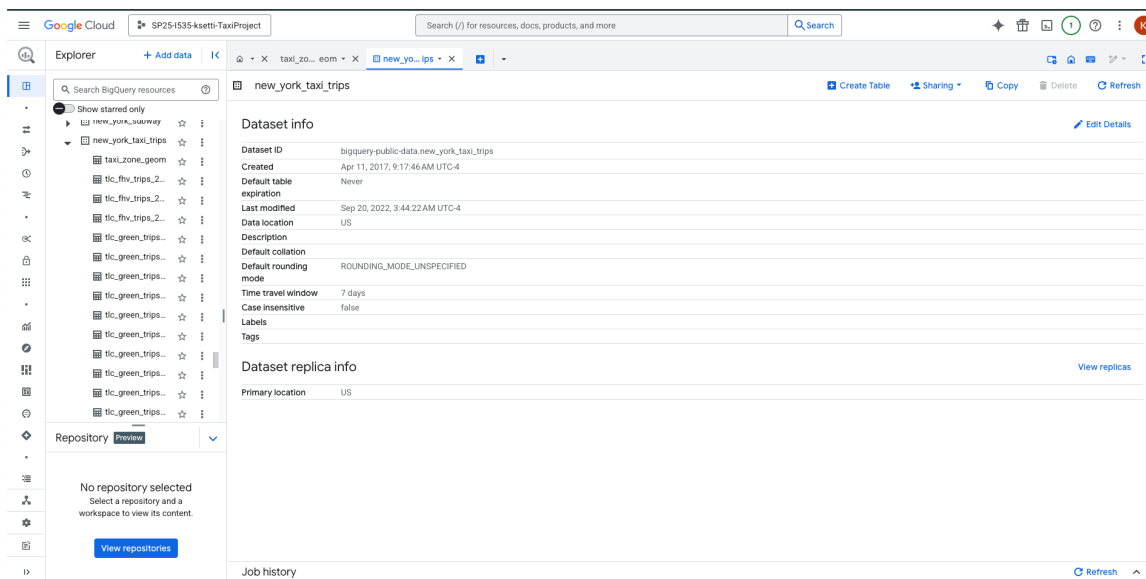


Figure 1: Sample preview of the raw dataset in BigQuery

Data Cleaning and Filtering in SQL

Before jumping into analysis, I used SQL queries to clean up the data. This included removing any trips where the distance was zero or negative, where the fare or tip amount was invalid, or where any important fields were missing. These kinds of filters ensured that only real, useful data would be considered in further analysis.

In addition, I extracted fields like pickup_month using SQL's EXTRACT () function. This would help later when grouping and visualizing data over time.

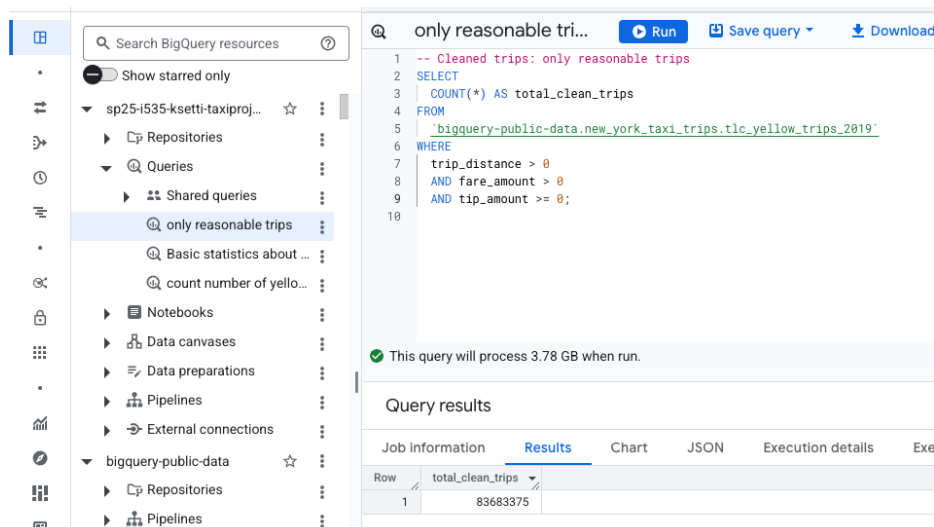


Figure 2: Query output showing total clean trip count

This step helped bring down the dataset from over 84 million records to a slightly smaller and more meaningful subset of clean trips. I also wrote queries to check some basic statistics such as total number of trips, and ensured the data was ready for grouping and analysis.

Data Visualization with Looker Studio

With clean data in hand, I connected BigQuery to Looker Studio to build interactive dashboards. Although the actual visual outputs are discussed in the Results section, the technical setup involved linking tables, creating date and zone dimensions, and configuring filters to support drilldowns in Looker.

This allowed for dynamic exploration of trip trends by time, geography, and payment type. No exports or manual uploads were needed, everything was kept within the cloud for speed and reproducibility.

PySpark Processing on Google Dataproc

To demonstrate distributed processing, I created a cluster using Google Cloud Dataproc (n4-standard-4 machine type with 4 vCPUs and 16 GB memory) and ran a PySpark job on a filtered January 2019 sample. The goal was to perform aggregations like average trip distance and tip amounts grouped by pickup zones.

The PySpark script was submitted as a job via the Dataproc UI. I used the Jupyter interface and Dataproc Notebook integration to preview results and validate outputs.

Cluster details

Consider using Auto Zone rather than selecting a zone manually. See <https://cloud.google.com/dataproc/docs/concepts/config>

Name	nyc-taxi-cluster
Cluster UUID	4e0b0088-2890-4ce4-b07f-d4a6748b4a37
Type	Dataproc Cluster
Status	Running

MONITORING JOBS VM INSTANCES **CONFIGURATION** WEB INTERFACES

EDIT

Region	us-central1
Zone	us-central1-c
Image version	2.2.54-debian12
Autoscaling	Off
Performance Enhancements	
Advanced optimizations	Off
Advanced execution layer	Off
Google Cloud Storage caching	Off
Dataproc Metastore	None
Scheduled deletion	Off
Confidential computing enabled?	Disabled
Master node	Single Node (1 master, 0 workers)
Machine type	n4-standard-2
Number of GPUs	0
Primary disk type	hyperdisk-balanced
Primary disk size	100GB
Local SSDs	0
Secure Boot	Disabled
VTPM	Disabled
Integrity Monitoring	Disabled
Cloud Storage staging bucket	dataproc-staging-us-central1-259857686910-xppcer9l
Network	default
Network tags	None
Internal IP only	Yes
Project access	Allow API access to all Google Cloud services in the same project
Created	May 1, 2025, 12:11:11 AM
Optional components	JUPYTER

Figure 3: Dataproc VM and cluster configuration used for PySpark processing

The visual output and insights generated from the PySpark run are shared in the Results section.

Results

This section presents the key findings from BigQuery SQL queries, Looker Studio visualizations, and PySpark analysis on Dataproc. Each insight builds on earlier steps and shows how different tools helped bring structure and meaning to the dataset.

- Total Number of Trips and Basic Stats:

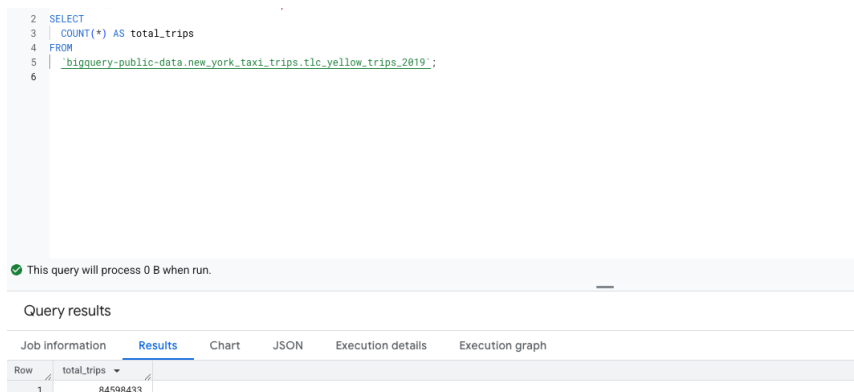


Figure 4: Count of total valid Yellow Taxi trips in 2019

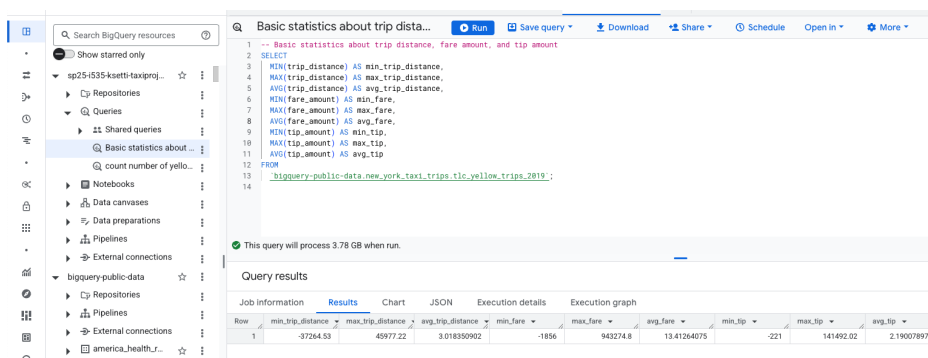


Figure 5: Query result showing average trip distance, fare amount, and tip amount

- Monthly and Hourly Trends:

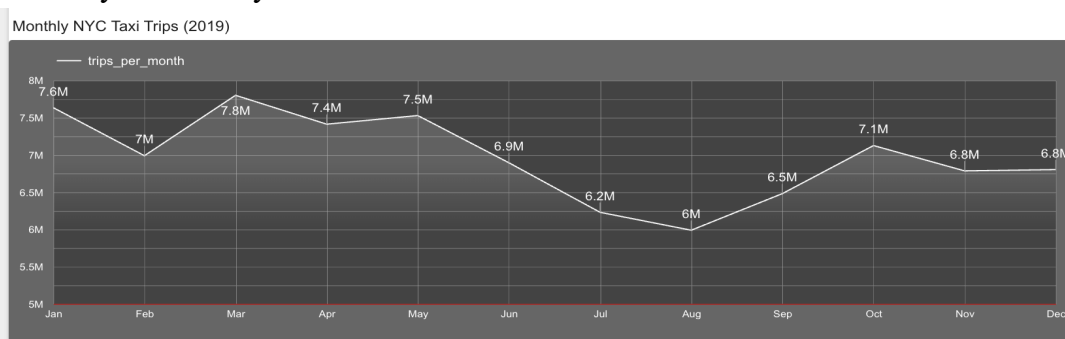


Figure 7: Monthly number of trips (Looker Studio line chart)

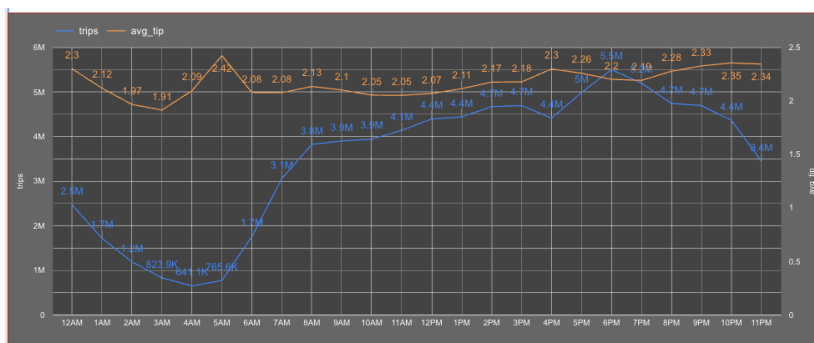


Figure 8: Hourly ride distribution and average tips

- Geographic Patterns:

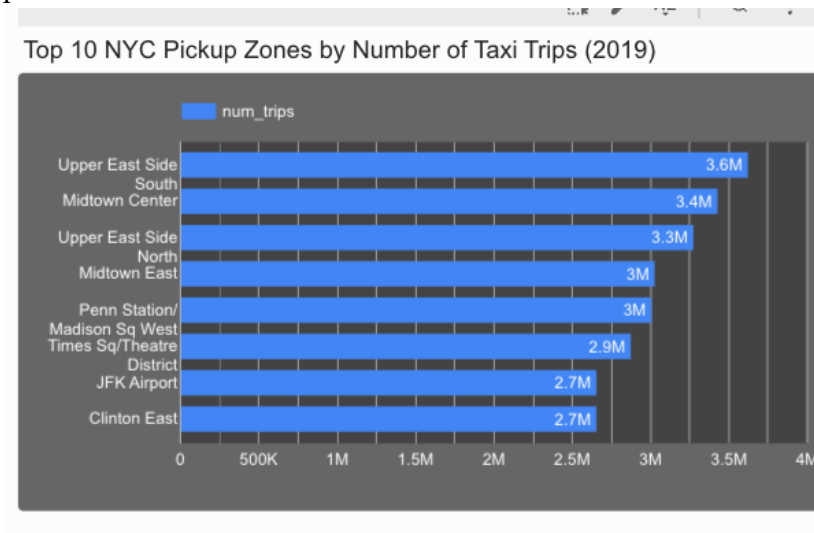


Figure 9: Top 10 pickup zones with real zone names

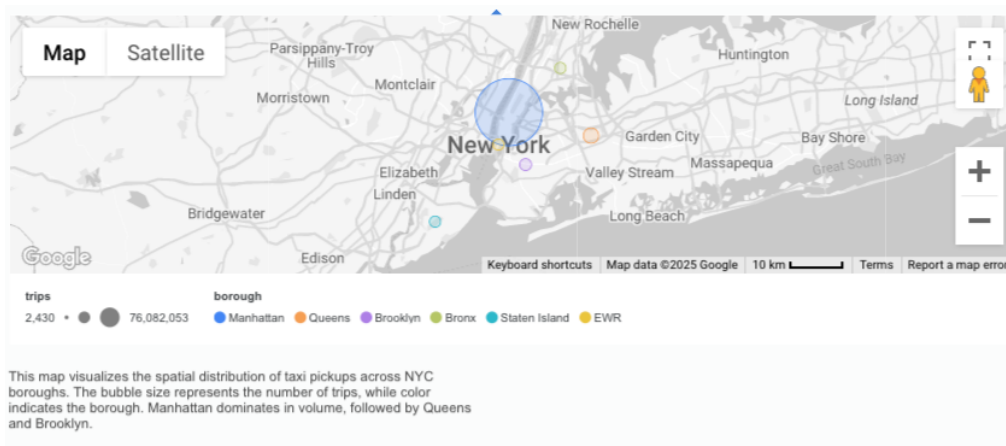


Figure 10: Map view of trip counts by borough or zone

- PySpark Aggregations on Dataproc:

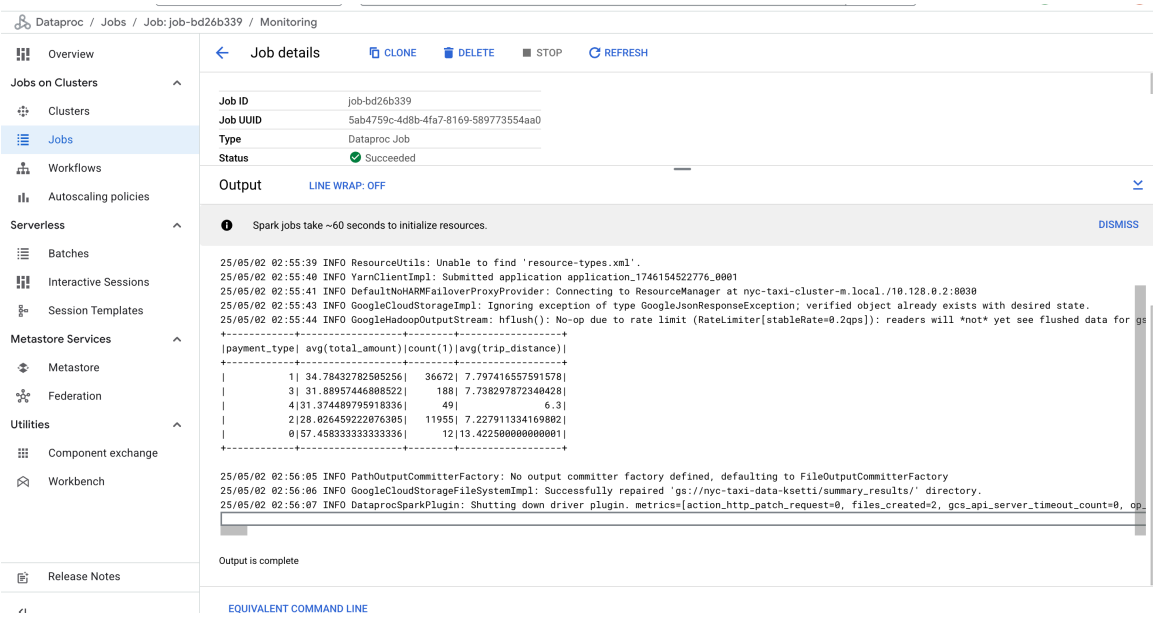


Figure 11: Dataproc output table showing average distance and tip by zone

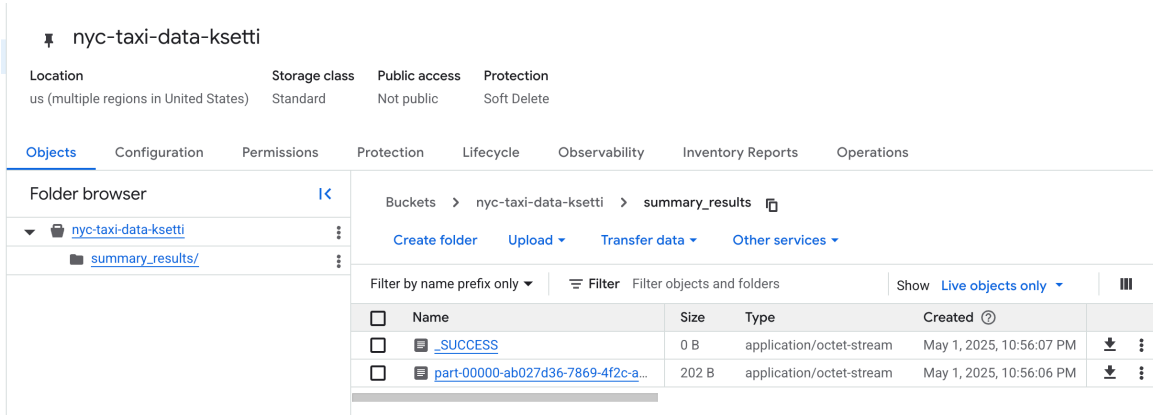


Figure 12: GCS bucket containing saved results from the PySpark job

Discussion

This project demonstrated the use of distributed cloud tools to build an end-to-end data pipeline for analyzing urban transportation data. Tools such as BigQuery, Dataproc, and Looker Studio were employed to query, clean, process, and visualize data efficiently at scale.

The project began by reviewing and querying a large dataset hosted on BigQuery. SQL was used to filter the data and calculate summary statistics. This provided early insights into total trip volumes, trip patterns, and anomalies in the data. Looker Studio visualizations allowed for a clear view of monthly ride trends and top pickup zones. The visual output enabled interactive exploration and interpretation, which added a layer of storytelling to the raw data.

In the next phase, Google Cloud Dataproc was used to showcase distributed data processing. Although the initial intention was to process the entire 2019 dataset, even limited processing using the smaller n4-standard-1 instance failed due to resource limitations. This highlighted the scale of the data and the necessity for more capable infrastructure. The project was then shifted to an n4-standard-4 instance, which was able to execute the PySpark job on the January 2019 subset. This sample allowed the pipeline to be tested and validated without hitting resource ceilings.

Several technical challenges arose during the process. Early efforts to configure PySpark locally hit roadblocks due to Java runtime issues and system incompatibility. Moving to GCP introduced its own difficulties — from VPC restrictions under the IU organization to firewall misconfigurations that blocked job execution. Resolving these issues involved manually adjusting network rules and port access settings, all of which helped in building a more in-depth understanding of how distributed systems operate behind the scenes.

Additional confusion was caused by a misleading error related to billing, which in reality was due to a typo in the bucket name. These setbacks, while frustrating, mirrored real-world scenarios where minor oversights can cause disproportionately large issues in cloud-based workflows. Despite this, the experience was valuable in teaching how to methodically troubleshoot and overcome such problems.

The project makes clear connections to the module covered in the course, particularly "Processing and Analytics". The use of BigQuery falls under the analytics and querying domain, while the shift to Dataproc and GCS for distributed processing and storage aligns well with lifecycle and pipeline design principles.

From a learning perspective, this project significantly enhanced my practical understanding of orchestrating big data workflows. It was not just about writing code but about stitching together different systems and making decisions on infrastructure, data handling, and optimization. It showed that big data is not just big because of its volume, it is big in complexity, in coordination, and in how it tests your problem-solving abilities.

Looker Studio also played a significant role in the final output. Unlike static visualizations, it allowed for an interactive exploration of the trends uncovered during the analysis. I created a multi-page dashboard to showcase trip patterns across months, identify top-performing pickup zones, and explore geographical insights visually. Its ease of integration with BigQuery and customization options made the entire visualization process smooth and scalable. Being able to use filters and drill-downs in the dashboard gave the analysis a professional and business-ready feel. This added a layer of clarity and storytelling that would not have been possible with tabular summaries alone.

Finally, to promote reproducibility and transparency, the dataset and relevant code used throughout the project have been published in a GitHub repository. This includes query scripts, transformation logic, and visualization snapshots. Publishing this work not only aligns with open data practices but also provides a reference for anyone interested in exploring or building upon the NYC taxi dataset using similar cloud-based tools.

Conclusion

This project brought together key ideas from the course, including distributed processing, pipeline design, cloud storage, and visualization, and applied them to real-world taxi data. From querying millions of records in BigQuery to processing samples with PySpark on Dataproc and building an interactive Looker Studio dashboard, each component illustrated how big data can be effectively handled in a modern cloud ecosystem. The challenges faced, especially during GCP setup and cluster execution, reinforced how complex such workflows can be and how important attention to detail becomes when working at scale. Overall, the experience helped strengthen my confidence in designing and managing big data systems

References

1. BigQuery NYC Taxi Dataset [link](#)
2. Google Cloud Dataproc [link](#)
3. Dataproc cluster network configuration [link](#)
4. Looker Studio [link](#)
5. PySpark Documentation [link](#)
6. GCP Storage Docs [link](#)
7. Github [link](#)